

Introduction to Statistics and Probability

WiMLDS Accra and AIMS Mathematics Bootcamp

AIMS Ghana

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WiMLDS
Women in Machine Learning & Data Science

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Overview

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2. Purpose and Learning Outcomes
3. Measures of Central Tendencies
4. Discrete & Continuous Probability Distributions
5. Hypothesis Testing (Test of Significance)
6. Exploratory Data Analysis(EDA)

Introduction

Introduction

We introduce two important concepts: Probability and Statistics.

- Statistics is the discipline that deals with the collection, organization, analysis, interpretation, and presentation of data.
- The term statistics can also be used to describe facts or summaries about data.
- Probability is the measure of how likely an event is to happen. It is expressed as a number between 0 and 1, where:
 1. Probability models are used to analyze trends and predict stock price movements, estimate risks and set premium rates, predict the likelihood of rain, storms, or temperature changes.
 2. The formula for finding probability is given as

$$P(E) = \frac{\text{Number of Favorable Outcomes}}{\text{Total Number of Outcomes}}$$

Purpose and Learning Outcomes

- The purpose of this presentation is to equip students with basic ideas of statistics and probability, their use and applications.
- By the end of this lesson, students should be able to understand;

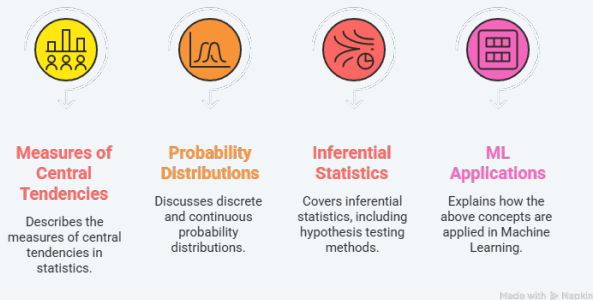
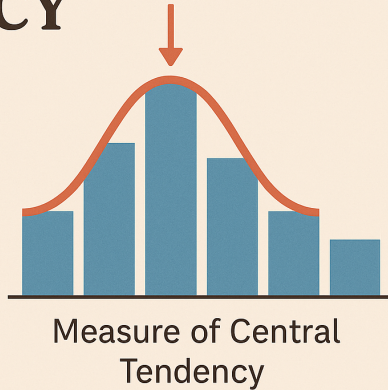


Figure: Learning Objectives

Measures of Central Tendencies

MEASURE OF CENTRAL TENDENCY

The representative value of a data set, generally the central value or the most occurring value that gives a general idea of the whole data set is called Measure of Central Tendency.



Mean

- The **population mean** of m numbers $x_1, x_2, \dots, x_m$ (the data for every member of a population of size m) is denoted by μ and is computed as follows:

$$\mu = \frac{x_1 + x_2 + \dots + x_m}{m}$$

- The **sample mean** of n numbers $x_1, x_2, \dots, x_n$ (the data for every member of a sample of size n - subset of the population) is denoted by $\bar{x}$ and is computed as follows:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

- Example:** Consider the following set of data, showing the number of times a sample of 5 students check their Tiktok accounts in a day: 9, 11, 13, 14, 15.
- Calculate the sample mean $\bar{x}$.

Mean Cont:

Mean of 5 students is given by

$$\begin{aligned}\bar{x} &= \frac{9 + 11 + 13 + 14 + 15}{5} \\ &= \frac{62}{5} = 12.4\end{aligned}$$

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Now imagine a Tiktok influencer who views their account 100 times a day joined the class. What is the new mean?

Mean Cont:

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Now imagine a Tiktok influencer who views their account 100 times a day joined the class. What is the new mean?

$$\begin{aligned}\bar{x} &= \frac{9 + 11 + 13 + 14 + 15 + 100}{6} \\ &= \frac{162}{6} = 27\end{aligned}$$

The Median

- **The Median** of a set of quantitative data is the middle number when the measurements are arranged in ascending order.
- **To Calculate the Median:** Arrange the n measurements in ascending (lowest to highest) order. We denote the median of the data by M .
 1. If n is odd, M is the middle number.
 2. If n is even, the median (M) is the average of the two middle numbers in the ordered dataset.
- **Example** The number of goals scored by the 32 teams in the 2014 world cup are shown below:
4, 6, 2, 4, 7, 3, 5, 2, 3, 10, 1, 4, 15, 1, 6, 3, 4, 8, 4, 5, 12, 18, 7, 1, 5, 11, 6, 2, 2, 3, 4, 3, 1
- Find the median of the above set of data.
- 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 3, 3, 4, 4, 4, 4, 4, 4, 5, 5, 5, 6, 6, 6, 7, 7, 8, 10, 11, 12, 15, 18
- **Answer: 4**

The Mode

- The **mode** of a set of measurements is the most frequently occurring value; it is the value having the highest frequency among the measurements.
- Example:** What is the mode of the data on the number of goals scored by each team in the world cup of 2006?
- 18, 15, 12, 11, 10, 8, 7, 7, 6, 6, 6, 5, 5, 5, 4, 4, 4, 4, 4, 4, 3, 3, 3, 3, 3, 2, 2, 2, 2, 1, 1, 1
1 (4) 2 (4) 3 (5) 4 (6)
5 (3) 6 (3) 7 (2) 8 (1)
10 (1) 11 (1) 12 (1) 15 (1)
18 (1)
- Answer: 4

Skewness

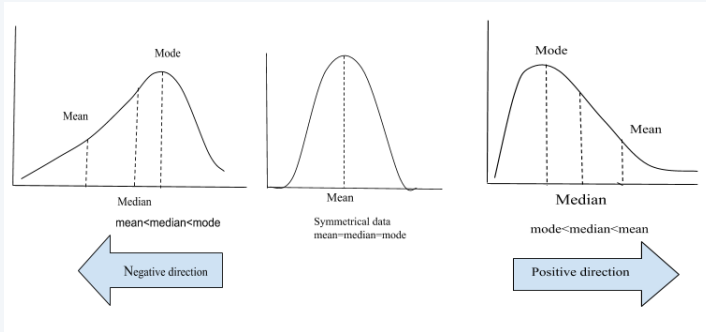


Figure: Different distributions and their skewness

Discrete & Continuous Probability Distributions

- **A Random variable** is a real-valued function that assigns values to each possible outcome of an experiment.
- The two types are: **Discrete** & **Continuous** random variables.
- **Discrete** random variables take on finite or countably infinite number of values. E.g, the number of Heads obtain from the toss of a coin twice.
- **Continuous** random variables take on infinite or uncountable finite number of values. E.g interval of time it takes for a code to run.
- **Probability Distribution** is the tabular or functional representation of a random variable and its respective probabilities.

Discrete Probability Distributions

1. Bernoulli Distribution
2. Binomial Distribution
3. Geometric Distribution
4. Negative Binomial(Pascal) Distribution
5. Poisson Distribution
6. Hypergeometric Distribution

Discrete Probability Distributions Cont:

- **The Bernoulli Distribution**

1. has two possible outcomes(success or failure).
2. single performance of the experiment.

$$P_X(X = x) = \begin{cases} p^x q^{1-x}, & \text{if } x = 0, 1 \\ 0, & \text{else} \end{cases}$$

- **The Binomial Distribution.**

1. n fixed number of trial
2. n independent Bernoulli Trials
3. two possible outcomes
4. constant success and failure probabilities.

$$P_X(X = x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & \text{if } x = 0, 1, \dots, n \\ 0, & \text{else} \end{cases}$$

Discrete Probability Distributions Cont:

- **Geometric Distribution**

$$P_X(X = x) = \begin{cases} pq^{x-1}, & \text{if } x = 1, 2, \dots \\ 0, & \text{else} \end{cases}$$

1. $E[X] = \frac{1}{p}$
2. $V[X] = \frac{q}{p^2}$

- **Negative Binomial Distribution**

$$P_X(X = x) = \begin{cases} \binom{x-1}{k-1} p^k q^{x-k}, & \text{if } x = k, k+1, \dots \\ 0, & \text{else} \end{cases}$$

- $E[X] = \frac{k}{p}$
- $V[X] = \frac{kq}{p^2}$

Continuous Probability Distributions

1. Normal Distribution
2. Exponential Distribution
3. Uniform Distribution
4. Gamma Distribution

Continuous Probability Distributions

- **Uniform Distribution**($X \sim U(a,b)$)

$$f_X(x) = \begin{cases} \frac{1}{b-a}, & \text{if } a \leq x \leq b, \\ 0, & \text{else} \end{cases}$$

- $E[X] = \frac{a+b}{2}$
- $V[X] = \frac{(b-a)^2}{12}$
- **Exponential Distribution**($X \sim \exp(\lambda)$)

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & \text{if } x \geq 0 \\ 0, & \text{else} \end{cases}$$

- $E[X] = \frac{1}{\lambda}$
- $V[X] = \frac{1}{\lambda^2}$

Continuous Probability Distributions Cont:

- **Normal Distribution($X \sim N(\mu, \sigma)$)**

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < X < \infty, -\infty < \mu < \infty, \sigma > 0$$

Hypothesis Testing (Test of Significance)

- Standard procedure, based on sample evidence and probability, used to test claims regarding a characteristic of one or more populations.

	True state of affairs (population)	
Decision	The null hypothesis is true H_0 is TRUE	The null hypothesis is false H_0 is not TRUE
We decide to reject the null hypothesis [reject H_0]	Type I error	Correct Decision
We fail to reject the null hypothesis [Do not Reject H_0]	Correct Decision	Type II error

Hypothesis Testing (Test of Significance Cont:)

- **Null Hypothesis(H_0):** This is the starting assumption. For example, "The average visits are 50."
- **Alternative hypothesis(H_A):** The opposite, saying there is a difference. For example, "The average visits are not 50."
- **Type I error(α):** When we reject the null hypothesis although that hypothesis was true.

$$\alpha = P(\text{reject } H_0 | H_0 \text{ is true})$$

- **Type II error (β):** When we accept the null hypothesis but it is false.

$$\beta = P(\text{Fail to reject } H_0 | H_0 \text{ is false})$$

- The two probabilities are inversely related.

Steps In Hypothesis Testing

- Choose the parameter of interest
- Specify/state the Hypothesis
- Specify the level of significance of the test
- Specify the appropriate Test Statistic and its Sampling Distribution
- Determine Statistical Significance
- Compute the Test Statistic
- Make Decision Rules
- Make a Statistical

Hypothesis Testing For Single Population Mean

- **Large Sample ($n \geq 30$) Assumptions**

1. Population is normally distributed
2. A random sample of size n is selected
3. If not normal, requires large samples. When n is large by C.L.T. the population has an approximate normal distribution.

- (a) $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$ (Two-Tailed alternative hypothesis)

1. If the population variance is known, we use z as the test statistic.
2. **Test Statistic:** $Z = \frac{\bar{X} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$
3. If the assumptions are correct and H_0 is true, the test statistic follows the standard normal distribution. Therefore, we calculate a z -score and use it to test the hypothesis.

Critical region is $\{Z : Z > z_{\frac{\alpha}{2}} \text{ or } Z < -z_{\frac{\alpha}{2}}\}$

Hypothesis Testing For Single Population Mean

Cont:

Decision:

- if observed value of the test statistic is in the critical region: **"Reject H_0 "**
- If observed value of the test statistic is not in the critical region:
"Do not Reject H_0 "

Conclusion: At 100% significance level there is (in)sufficient statistical evidence to
"favor H_1 "

Exploratory Data Analysis(EDA)

- An analysis approach that identifies general patterns in the data. The steps are;
- **Data Collection:** Gather the dataset you want to analyze. This can be from various sources such as databases, surveys, APIs, etc.
- **Data Cleaning:** Clean the dataset by handling missing values, outliers, and inconsistencies. This step ensures that the data is suitable for analysis.
- **Univariate Analysis:** Explore individual variables in the dataset to understand their distributions, central tendency, spread, and detect outliers. Common techniques include histograms, box plots, and summary statistics.
- **Bivariate Analysis:** Examine the relationship between pairs of variables in the dataset. This can involve scatter plots, correlation analysis, and contingency tables.

- **Multivariate Analysis:** Analyze relationships between multiple variables simultaneously. Techniques such as heatmaps, pair plots, and cluster analysis can be used to identify patterns and correlations among variables.
- **Visualization:** Create visualizations to summarize and present findings from the analysis effectively. This can include bar charts, pie charts, line plots, and more complex visualizations like heatmaps and network diagrams.
- **Statistical Testing:** Conduct hypothesis tests and statistical inference to validate findings and make predictions about the population based on the sample data.
- **Iteration:** EDA is often an iterative process. As you gain insights from the initial analysis, you may go back to earlier steps to refine your understanding or explore new questions.

Histogram

- A histogram summarizes the distribution of the data by placing observations into intervals (also called classes or bins) and counting the number of observations in each interval.

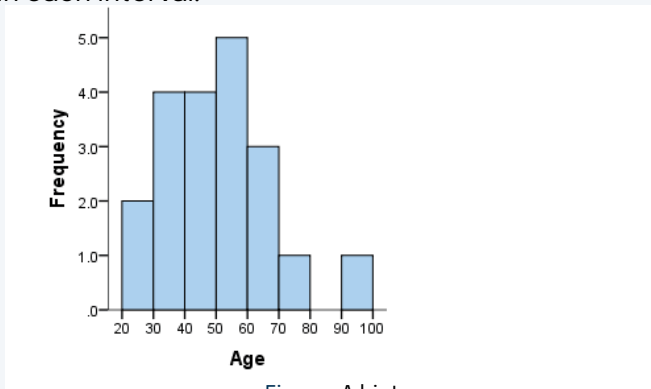


Figure: A histogram

Boxplot

- A box and whisker plot (also referred to as boxplot) provides a compact summary of the distribution of a variable.
- Used for numerical data

Boxplot Cont:

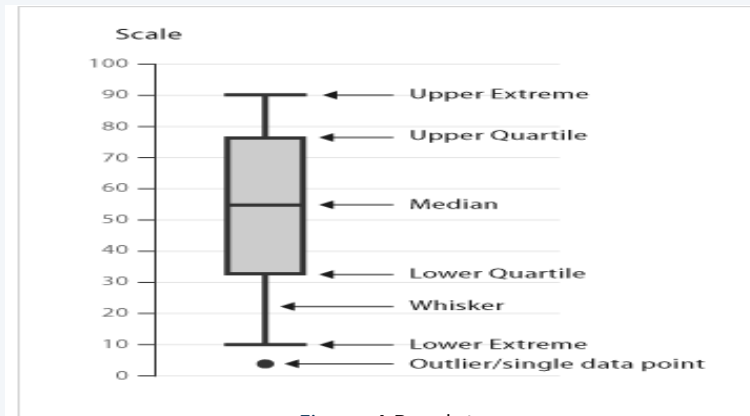


Figure: A Boxplot

Barplot

- A barplot is a type of data visualization that represents categorical data with rectangular bars.
- The length of each bar corresponds to the value it represents.
- Barplots are commonly used to compare the values of different categories or to show the distribution of a single categorical variable.

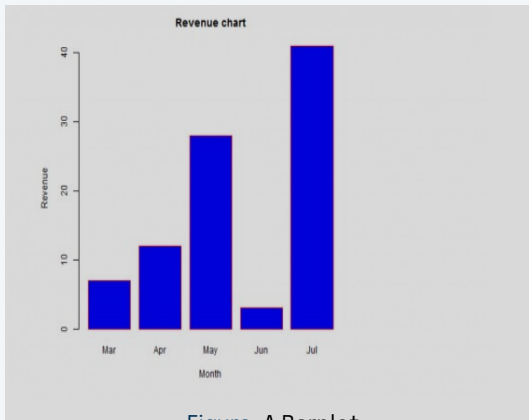


Figure: A Barplot

Pie Chart

- A pie chart is another type of data visualization used to represent categorical data.
- Unlike a barplot, which uses rectangular bars, a pie chart uses a circular shape to represent data.
- The entire circle represents the total sum of the data, and each category is represented by a slice of the pie.

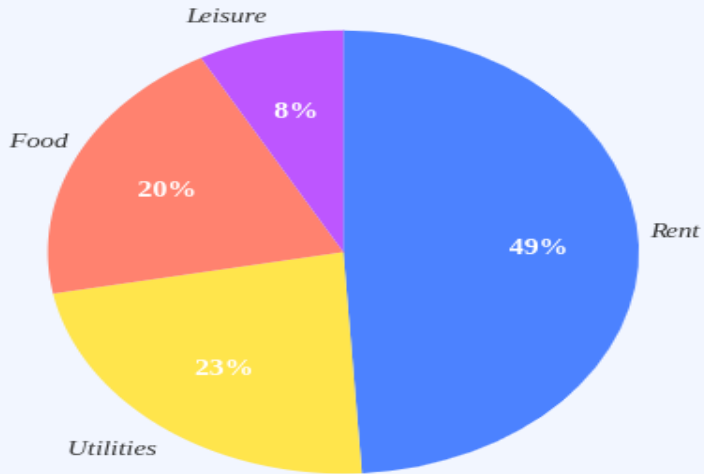


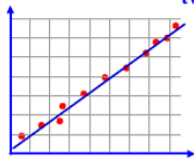
Figure: A Pie Chart

Scatter Plot and Correlation

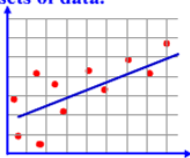
- A scatter plot is a type of data visualization that is used to display the relationship between two continuous variables.
- Each point on the plot represents an observation in the dataset, with the x-coordinate representing one variable and the y-coordinate representing the other variable.

SCATTERPLOTS & CORRELATION

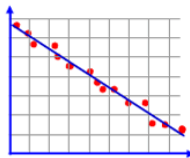
Correlation - indicates a relationship (connection) between two sets of data.



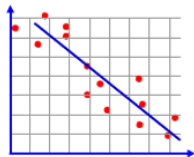
Strong positive correlation



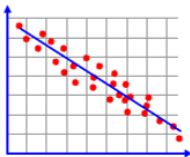
Weak positive correlation



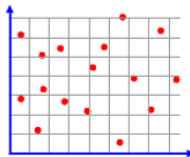
Strong negative correlation



Weak negative correlation



Moderate negative correlation



No correlation

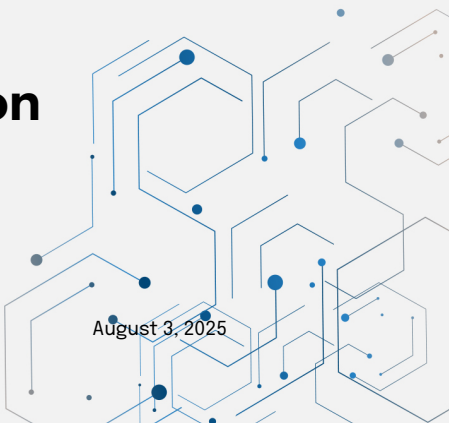
Figure: Scatter plot and correlation



Thank you for your attention

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